Seed-count stability analysis

Partly. The secondary fixed-split benefits become smaller with ten seeds, especially NH3. The primary multi-fold results do not weaken systematically: NH3/H2 improve slightly, while CO/ethylene attenuate slightly. Drift partition 8 weakens, partition 9 is stable, and partition 10 has no clear ten-seed transfer benefit.

These comparisons use prefixes of the same corrected protocol in the predefined order 42, 142, …, 942. They do not compare a favorable selected three-seed subset against ten seeds. Positive percentages mean reduced MAE versus scratch. All ten seeds remain in the final report.

Evaluation	Gas	First 3 reduction	All 10 reduction	Later 7 reduction
primary_multifold	CO	28.48%	27.50%	27.06%
primary_multifold	Ethylene	26.09%	25.73%	25.57%
primary_multifold	H2	18.40%	21.49%	22.72%
primary_multifold	NH3	13.30%	13.91%	14.15%
fixed_split	CO	27.23%	25.43%	24.71%
fixed_split	Ethylene	31.14%	24.31%	21.01%
fixed_split	H2	27.22%	23.29%	21.52%
fixed_split	NH3	32.44%	19.59%	12.96%
drift_partition_8	MacroMAE	25.61%	18.80%	15.63%
drift_partition_9	MacroMAE	17.97%	18.35%	18.52%
drift_partition_10	MacroMAE	-11.95%	-0.50%	4.17%

Primary uncertainty after ten seeds

Gas	Paired MAE reduction (ppm)	95% seed/fold interval (ppm)
CO	24.737	[12.850, 37.527]
Ethylene	0.672	[0.247, 1.143]
H2	10.575	[3.566, 18.776]
NH3	4.312	[0.381, 8.305]

All four primary intervals exclude zero, with NH3 closest to zero. They resample paired seeds and folds, condition on the observed exposures, and do not create independent acquisition evidence. The full-prefix CSV and random-order envelope show variability at every count; the envelope is an order/subset diagnostic, not a population confidence interval.

Comparison with the historical manuscript

Gas	Historical 3-repeat reduction	Revised 10-seed reduction
CO	20.41%	27.50%
Ethylene	14.55%	25.73%
H2	20.11%	21.49%
NH3	16.39%	13.91%

This historical comparison is not a seed-only experiment. The revision changes policy reporting, uses matched fitting/validation holdouts, and uses full available adaptation coverage in rolling folds. Thus the same-protocol prefix comparison above is the appropriate test of adding seeds. More seeds characterize initialization, training, calibration-design and holdout variability; they cannot compensate for a small number of independently acquired test partitions.

